Insu Yun

Associate Professor (Untenured)
School of Electrical Engineering,
Korea Advanced Institute of Science and Technology (KAIST)
Email: insuyun@kaist.ac.kr
Web: https://insuyun.github.io

Research Interests

System security, software security, binary analysis, fuzzing, and applied cryptography.

Education

Georgia Institute of Technology Aug. 2015 – Dec. 2020

Ph.D. in Computer Science

Advisor: Dr. Taesoo Kim

Korea Advanced Institute of Science and Technology (KAIST) Sep. 2008 – Feb. 2015

B.S. in Computer Science & Mathematics

Work Experience

KAIST, Daejeon, South Korea Feb. 2021 –

Associate Professor Mar. 2025 – Present

Assistant Professor Feb. 2021 – Feb. 2025

Microsoft Research, Research Intern, Seattle, WA May. 2017 – Aug. 2017

Contributed to REPT, a system that utilizes Intel Processor Trace to diagnose production failures

Mentor: Weidong Cui

Georgia Tech, Research Assistant, Atlanta, GA Aug. 2015 – Dec. 2020

Korean Cyber Command, Software Developer, Seoul, Korea Apr. 2012 – Jan. 2014

Served for the mandatory military service

Publications

* Students advised by myself are underlined.

International Conferences (Top-tier and others)

27. **Function-Level Fuzzing for RTOS Kernels with RTCon**
    Insu Yun, and Eunkyu Lee
    Proceedings of the 2026 IEEE International Conference on Software Testing, Verification and Validation Tutorial (ICST 2026)
    Daejeon, Republic of Korea, May 2026

26. **RTCon: Context-Adaptive Function-Level Fuzzing for RTOS Kernels**
    Eunkyu Lee, Junyoung Park, and **Insu Yun**
    Proceedings of the 2026 Annual Network and Distributed System Security Symposium (NDSS 2026)
    February 2026

25. **OTABase: Enhancing Over-the-Air Testing to Detect Memory Crashes in Cellular Basebands**
    CheolJun Park, Marc Egli, BeomSeok Oh, Tuan Dinh Hoang, Suhwan Jeong, Martin Crettol, **Insu Yun**, Mathias Payer, and Yongdae Kim

Proceedings of the 41st Annual Computer Security Applications Conference (ACSAC 2025)
Austin, TX, December 2025

24. **CROSS-X: Generalized and Stable Cross-Cache Attack on the Linux Kernel**
Dongok Kim*, Juhyun Song*, and **Insu Yun**
Proceedings of the 32nd ACM Conference on Computer and Communications Security (CCS 2025)
Taipei, Taiwan, October 2025

23. **Windows plays Jenga: Uncovering Design Weaknesses in Windows File System Security**
Dong-uk Kim*, Junyoung Park*, Sanghak Oh, Hyoungshick Kim, and **Insu Yun**
Proceedings of the 32nd ACM Conference on Computer and Communications Security (CCS 2025)
Taipei, Taiwan, October 2025

22. **LLFuzz: An Over-the-Air Dynamic Testing Framework for Cellular Baseband Lower Layers**
Tuan Dinh Hoang, Taekkyung Oh, CheolJun Park, **Insu Yun**, and Yongdae Kim
Proceedings of the 34th USENIX Security Symposium (Security 2025)
Seattle, WA, August 2025

21. **Too Much of a Good Thing: (In-)Security of Mandatory Security Software for Financial Services in South Korea**
Taisic Yun, Suhwan Jeong, Yonghwa Lee, Seungjoo Kim, Hyoungshick Kim, **Insu Yun**, and Yongdae Kim
Proceedings of the 34th USENIX Security Symposium (Security 2025)
Seattle, WA, August 2025

20. **FirmState: Bringing Cellular Protocol States to Shannon Baseband Emulation**
Suhwan Jeong, Beomseok Oh, Kwangmin Kim, **Insu Yun**, Yongdae Kim, and CheolJun Park
Proceedings of the 18th ACM Conference on Security and Privacy in Wireless and Mobile Networks (WiSec 2025)
Arlington, VA, June 2025

19. **Automated Attack Synthesis for Constant Product Market Makers**
Sujin Han, Jinseo Kim, Sung-Ju Lee, and **Insu Yun**
Proceedings of the ACM SIGSOFT International Symposium on Software Testing and Analysis 2025 (ISSTA 2025)
Trondheim, Norway, June 2025

18. **Bridging the Gap between Real-World and Formal Binary Lifting through Filtered-Simulation**
Jihee Park, **Insu Yun**, and Sukyoung Ryu
Proceedings of the ACM SIGPLAN International Conference on Object-Oriented Programming, Systems, Languages, and Applications 2025 (OOPSLA 2025)
Singapore, October 2025

17. **RGFuzz: Rule-Guided Fuzzer for WebAssembly Runtimes**
Junyoung Park, Yunho Kim, and **Insu Yun**
Proceedings of the 46th IEEE Symposium on Security and Privacy (Oakland 2025)
San Francisco, CA, May 2025

16. **BaseComp: A Comparative Analysis for Integrity Protection in Cellular Baseband Software**
Eunsoo Kim*, Min Woo Baek*, CheolJun Park, Dongkwan Kim, Yongdae Kim, and **Insu Yun**
Proceedings of the 32nd USENIX Security Symposium (Security 2023)
Anaheim, CA, August 2023

15. **QueryX: Symbolic Query on Decompiled Code for Finding Bugs in COTS Binaries**
HyungSeok Han, JeongOh Kyea, Yonghwi Jin, Jinoh Kang, Brian Park, and **Insu Yun**
Proceedings of the 44th IEEE Symposium on Security and Privacy (Oakland 2023)
San Francisco, CA, May 2023

14. **Fuzzing@Home: Distributed Fuzzing on Untrusted Heterogeneous Clients**
    Daehee Jang, Ammar Askar, **Insu Yun**, Stephen Tong, Yiqin Cai, and Taesoo Kim
    Proceedings of the 2022 International Symposium on Research in Attacks, Intrusions and Defenses (RAID 2022)
    October 2022

13. **DoLTEst: In-depth Downlink Negative Testing Framework for LTE Devices**
    CheolJun Park*, Sangwook Bae*, BeomSeok Oh, Jiho Lee, Eunkyu Lee, **Insu Yun**, and Yongdae Kim
    Proceedings of the 31th USENIX Security Symposium (Security 2022)
    Boston, MA, August 2022
    (Acceptance rates: 18%, 256/1414)

12. **HardsHeap: A Universal and Extensible Framework for Evaluating Secure Allocators**
    Insu Yun, Woosun Song, Seunggi Min, and Taesoo Kim
    Proceedings of the 28th ACM Conference on Computer and Communications Security (CCS 2021)
    Seoul, South Korea, November 2021
    (Acceptance rates: 22%, 196/880)

11. **Preventing Use-After-Free Attacks with Fast Forward Allocation**
    Brian Wickman, Hong Hu, **Insu Yun**, Daehee Jang, JungWon Lim, Sanidhya Kashyap, and Taesoo Kim
    Proceedings of the 30th USENIX Security Symposium (Security 2021)
    Vancouver, B.C., Canada, August 2021
    (Acceptance rates: 19%, 246/1316)

10. **BaseSpec: Comparative Analysis of Baseband Software and Cellular Specifications for L3 Protocols**
    Eunsoo Kim*, Dongkwan Kim*, Cheoljun Park, **Insu Yun**, and Yongdae Kim
    Proceedings of the 2021 Annual Network and Distributed System Security Symposium (NDSS 2021)
    February 2021
    (Acceptance rates: 15%, 87/578)

9. **Automatic Techniques to Systematically Discover New Heap Exploitation Primitives**
   Insu Yun, Dhaval Kapil, and Taesoo Kim
   Proceedings of the 29th USENIX Security Symposium (Security 2020)
   Boston, MA, August 2020
   (Acceptance rates: 16%, 157/977)

8. **Fuzzing JavaScript Engines with Aspect-preserving Mutation**
   Soyeon Park, Wen Xu, **Insu Yun**, Daehee Jang, and Taesoo Kim
   Proceedings of the 41st IEEE Symposium on Security and Privacy (Oakland 2020)
   San Francisco, CA, May 2020
   (Acceptance rates: 12%, 104/841)
   Nominated as a finalist in CSAW Best Applied Research Paper Award 2020

7. **REPT: Reverse Debugging of Failures in Deployed Software**
   Weidong Cui, Xinyang Ge, Baris Kasikci, Ben Niu, Upamanyu Sharma, Ruoyu Wang, and **Insu Yun** (alphabetical)
   Proceedings of the 13th USENIX Symposium on Operating Systems Design and Implementation (OSDI 2018)
   Carlsbad, CA, October 2018
   (Acceptance rates: 18%, 47/257)
   Jay Lepreau Best Paper Award (3 out of 257 submissions)

6. **QSYM: A Practical Concolic Execution Engine Tailored for Hybrid Fuzzing**
   Insu Yun, Sangho Lee, Meng Xu, Yeongjin Jang, and Taesoo Kim
   Proceedings of the 27th USENIX Security Symposium (Security 2018)
   Baltimore, MD, August 2018
   (Acceptance rates: 19%, 100/524)
   Distinguished Paper Award (5 out of 524 submissions)

5. **CAB-Fuzz: Practical Concolic Testing Techniques for COTS Operating Systems**
   Su Yong Kim, Sangho Lee, **Insu Yun**, Wen Xu, Byoungyoung Lee, Youngtae Yun, and Taesoo Kim
   Proceedings of the 2017 USENIX Annual Technical Conference (ATC 2017)
   Santa Clara, CA, July 2017
   (Acceptance rates: 21%, 60/283)

4. **APISan: Sanitizing API Usages through Semantic Cross-checking**
   Insu Yun, Changwoo Min, Xujie Si, Yeongjin Jang, Taesoo Kim, and Mayur Naik
   Proceedings of the 25th USENIX Security Symposium (Security 2016)
   Austin, TX, August 2016
   (Acceptance rates: 16%, 72/463)
   Nominated as a finalist in CSAW Best Applied Research Paper Award 2016

3. **HDFI: Hardware-Assisted Data-Fow Isolation**
   Chengyu Song, Hyungon Moon, Monjur Alam, **Insu Yun**, Byoungyoung Lee, Taesoo Kim, Wenke Lee, and Yunheung Paek
   Proceedings of the 37th IEEE Symposium on Security and Privacy (Oakland 2016)
   San Jose, CA, May 2016
   (Acceptance rates: 13%, 55/413)

2. **Analyzing Security of Korean USIM-based PKI Certificate Service**
   Shinjo Park, Suwan Park, **Insu Yun**, Dongkwan Kim, and Yongdae Kim
   Proceedings of the 15th International Workshop on Information Security Applications (WISA 2014)
   Jeju Island, Korea, August 2014

1. **Kargus: A Highly-scalable Software-based Intrusion Detection System**
   Muhammad Jamshed, Jihyung Lee, Sangwoo Moon, **Insu Yun**, Deokjin Kim, Sungryoul Lee, Yung Yi, and KyoungSoo Park
   Proceedings of the 19th ACM Conference on Computer and Communications Security (CCS 2012)
   Raleigh, NC, October 2012
   (Acceptance rates: 19%, 81/426)

International Journal

1. **Scalable and Secure Virtualization of HSM with ScaleTrust**
   Juhyeng Han, **Insu Yun**, Seongmin Kim, Taesoo Kim, Sooel Son, and Dongsu Han
   IEEE/ACM Transactions on Networking (ToN)
   November 2022

Other Refereed Materials

4. **From the Vulnerability to the Victory: A Chrome Renderer 1-Day Exploit's Journey to v8CTF Glory**
   <u>Haein Lee</u>, and **Insu Yun**
   TyphoonCon 2024
   Seoul, Korea, May 2024

3. **One shot, Triple kill: Pwning all three Google kernelCTF instances with a single 1-day Linux vulnerability**
   <u>Dongok Kim</u>, <u>Seunghyun Lee</u>, and **Insu Yun**
   POC 2023
   Seoul, Korea, November 2023

2. **Compromising the macOS kernel through Safari by chaining six vulnerabilities**
   Yonghwi Jin, Jungwon Lim, **Insu Yun**, and Taesoo Kim
   Black Hat USA Briefings (Black Hat USA 2020)
   Las Vegas, NV, August 2020

1. **AVPASS: Leaking and Bypassing Antivirus Detection Model Automatically**
   Jinho Jung, Chanil Jeon, Max Wolotsky, **Insu Yun**, and Taesoo Kim
   Black Hat USA Briefings (Black Hat USA 2017)
   Las Vegas, NV, July 2017

Thesis

1. **Concolic Execution Tailored for Hybrid Fuzzing**
   Insu Yun
   Ph.D. thesis, Georgia Institute of Technology
   Atlanta, GA, December 2020

Professional Activities

Technical Program Committee (International)

Program Committee, *IEEE Symposium on Security and Privacy (Oakland)*, 2024–2026

Program Committee, *Network and Distributed System Security Symposium (NDSS)*, 2024–2026

Program Committee, *The International Conference on Cryptology and Network Security (CANS)*, 2025

Program Committee, *ACM Conference on Security and Privacy in Wireless and Mobile Networks (WiSec)*, 2022–2023

Journal Editor (International)

Associate Editor, *ACM Transaction on Storage (ToS)*, 2024–2026

Others (International)

Organization Committee (Publicity Chair), *Network and Distributed System Security Symposium (NDSS)*, 2027

Organization Committee (Web Chair), *ACM Conference on Security and Privacy in Wireless and Mobile Networks (WiSec)*, 2024

Artifact Evaluation Committee, *ACM Conference on Computer and Communications Security (CCS)*, 2023

Artifact Evaluation Committee, *USENIX Security Symposium (Security)*, 2023

Organization Committee (Web Chair), *ACM Conference on Computer and Communications Security (CCS)*, 2021

Technical Program Committee (Domestic)

Program Committee, *Conference on Information Security and Cryptography Summer (CISC-S)*, 2026

Program Committee, *Conference on Information Security and Cryptography Winter (CISC-W)*, 2025

Program Committee, *Conference on Information Security and Cryptography Summer (CISC-S)*, 2021

Others (Domestic)

Policy Advisory Committee, *Ministry of National Defense*, 2026

Security Special Committee, *National Artificial Intelligence Strategy Committee*, 2026

Security Task Force, *National Artificial Intelligence Strategy Committee*, 2025

Advisory Boards, *AI Cyber Defense Contest (ACDC)*, 2025

Advisory Boards, *HackTheon Sejong*, 2024–2025

Organization Committee, *Conference on Information Security and Cryptography Summer (CISC-S)*, 2021

Teaching Experience

Advanced Programming Techniques for Electrical Engineering (EE309 at KAIST) — Fall 2025

- Evaluation – Average: 4.29 / 5

Software Hacking Theory and Practice (EE517 at KAIST) — Spring 2025

- Evaluation – Average: 4.67 / 5

Advanced Programming Techniques for Electrical Engineering (EE309 at KAIST) — Fall 2024

- Evaluation – Average: 4.54 / 5

Information Security Laboratory (IS521 at KAIST) Fall 2024

- Evaluation – Average: 4.64 / 5

Software Hacking Theory and Practice (EE517 at KAIST) Spring 2024

- Evaluation – Average: 4.92 / 5

Advanced Programming Techniques for Electrical Engineering (EE309 at KAIST) Fall 2023

- Evaluation – Average: 4.57 / 5

Software Hacking Theory and Practice (EE517 at KAIST) Spring 2023

- Evaluation – Average: 4.54 / 5

Programming Structures for Electronical Engineering (EE209 at KAIST) Fall 2022

- Evaluation – Average: 4.65 / 5

Software development environment and tools practice (EE485-A at KAIST) Fall 2022

- Evaluation – Average: 4.43 / 5

Software Security (EE595-B at KAIST) Spring 2022

- Evaluation – Average: 5 / 5

Programming Structures for Electronical Engineering (EE209 at KAIST) Fall 2021

- Evaluation – Average: 4.34 / 5

Software development environment and tools practice (EE485-A at KAIST) Fall 2021

- Evaluation – Average: 4.34 / 5

Software Security (EE595-B at KAIST) Spring 2021

- Evaluation – Average: 4.9 / 5

Honors & Awards

Academic awards

KAIST Q-Day Special Commendation for Research, KAIST Nov. 2025

Song-Am Future Researcher Award, KAIST Feb. 2025

Technology Innovation Awards, KAIST College of Engineering Dec. 2024

Best Teaching Award, KAIST Electrical Engineering Oct. 2024

Prize for Excellence in Teaching, KAIST Feb. 2024

Frontiers of Science Award, The First International Congress of Basic Science (ICBS) July. 2023

Best Teaching Award, KAIST Electrical Engineering Sep. 2021

Jay Lepreau Best Paper Award, USENIX OSDI 2018 Aug. 2018

Distinguished Paper Award, USENIX Security 2018 Aug. 2018

Hacking competitions

DEFCON 26 CTF, 1st place (Team DEFKOR00T) Aug. 2018

DEFCON 24 CTF, 3rd place (Team DEFKOR) Aug. 2016

DEFCON 23 CTF, 1st place (Team DEFKOR) Aug. 2015

Whitehat contest 2014, 1st place (Team SysSec) Nov. 2014

DEFCON 22 CTF, 10th place (Team GoN) Aug. 2014

SECCON CTF 2014, 1st place (TOEFL Beginner) Feb. 2014

Codegate CTF 2012, 3rd place (Team GoN) Apr. 2012

Secuinside CTF, 3rd place (Team GoN) Oct. 2011

ISEC CTF, 1st place (Team GoN) Sep. 2011

DEFCON 18 CTF, 3rd place (Team GoN) Aug. 2010

Codegate CTF 2010, 5th place (Team GoN) Apr. 2010

KISA HDCON, Gold Medal, 2nd place (Team GoN) May 2009

Codegate CTF 2009, 4th place (Team GoN) Apr. 2009

Scholarships

National Research Foundation of Korea Scholarship for Undergraduate — Mar. 2008 – Dec. 2013

Others

DARPA AIxCC, 1st place (Team Atlanta), $4M award — Aug. 2025

Cyber Security Challenge, 2nd place (Team HackingLab), $400K research grant — 2023

DARPA Cyber Grand Challenge, Finalist (Team Disekt) — Aug. 2016

KISA Bug Bounty Program's Hall of Fame — 2013

Vulnerability Discovery Reward (aka Bug bounty)

To summarize, $244.6K (by my students) and $92.8K (by myself) bug bounties are awarded so far.

By my students

Pwn2Own - Microsoft Edge and Google Chrome ($145K), ZDI, by SeunHyun Lee — Mar. 2024

v8CTF - CVE-2023-6702 ($10K), Google, by Haein Lee — Jan. 2024

kernelCTF - CVE-2023-3390 ($67.8K), Google, by Dongok Kim and SeunHyun Lee — Oct. 2023

Type confusion in V8 ($7K), Google, by Haein Lee — Mar. 2023

NAS authentication bypass in Exynos ($14.8K), Samsung Electronics, by Eunsoo Kim and CheolJun Park — Feb. 2022

By myself

PSV-2021-0304: afpd auth bypass ($300), NETGEAR — Mar. 2021

Pwn2Own Apple Safari with a kernel privilege escalation ($70K), ZDI, with Yonghwi Jin and Jungwon Lim — Mar. 2020

Apple Safari sandbox escape ($20K), Apple — Dec. 2019

Three integer overflow vulnerabilities in PHP ($1.5K), the Internet Bug Bounty — Jun. 2016

An Integer Overflow in Python zipimport ($1K), the Internet Bug Bounty — Apr. 2016

Patents

International

3. **Dynamic testing method and system for detecting memory corruptions in lower layers of cellular baseband of mobile communication devices (Pending)**
   Inventors: Yongdae Kim, Hoang Dinh Tuan, Insu Yun, Taekkyung Oh, Beomseok Oh, Mincheol Son, CheolJun Park
   Application date: 2026.01.07
   Application number: 19442669
   Country: US
2. **Security analysis system and method based on negative testing for protocol implementation of LTE device**
   Inventors: Yongdae Kim, CheolJun Park, Sangwook Bae, Beomseok Oh, Jiho Lee, Mincheol Son, Insu Yun
   Registration date: 12511402
   Patent number: 2025.12.30
   Country: US
1. **Reverse debugging of software failures**
   Inventors: Weidong Cui, Xinyang Ge, Baris Kasikci, Cengiz Can, Ben Niu, Ruoyu Wang, Insu Yun
   Registration date: 10565511

Patent number: 2020.02.18
Country: US

Domestic

9. **Method and apparatus for detecting memory vulnerabilities based on network-side state control in wireless communication systems (Pending)**
   Inventors: Yongdae Kim, Beomseok Oh, Insu Yun, Mincheol Son, Hoang Dinh Tuan, Soohwan Jeong, CheolJun Park
   Application date: 2025.12.03
   Application number: 10-2025-0189113
   Country: Korea

8. **Method and apparatus for fuzzing-based vulnerability detection (Pending)**
   Inventors: Insu Yun, Eunkyu Lee, Junyeong Park
   Application date: 2025.11.11
   Application number: 10-2025-0169729
   Country: Korea

7. **Dynamic testing method and system for detecting memory vulnerabilities in lower layers of cellular baseband of mobile communication devices (Pending)**
   Inventors: Yongdae Kim, Taekkyung Oh, Beomseok Oh, Mincheol Son, CheolJun Park, Hoang Dinh Tuan, Insu Yun
   Application date: 2025.05.07
   Application number: 10-2025-0059118
   Country: Korea

6. **Heuristic-based binary lifting and verification system (Pending)**
   Inventors: Sukyoung Ryu, Jihee Park, Insu Yun
   Application date: 2025.03.31
   Application number: 10-2025-0041131
   Country: Korea

5. **Method and apparatus for decompilation-specialized static binary translation**
   Inventors: Insu Yun, Woosun Song, Seunggi Min
   Registration date: 10-2834292-0000
   Patent number: 2025.07.10
   Country: Korea

4. **Security analysis system and method based on negative testing for protocol implementation of LTE device**
   Inventors: Yongdae Kim, CheolJun Park, Sangwook Bae, Beomseok Oh, Jiho Lee, Mincheol Son, Insu Yun
   Registration date: 10-2514797-0000
   Patent number: 2023.03.23
   Country: Korea

3. **Method and system for automatically analyzing bugs in cellular baseband software using comparative analysis based on cellular specifications**
   Inventors: Yongdae Kim, Eunsoo Kim, Dongkwan Kim, CheolJun Park, Insu Yun
   Registration date: 10-2546946-0000
   Patent number: 2023.06.20
   Country: Korea

2. **Methods and systems for key management service provision**
   Inventors: Dongsoo Han, JuHyeng Han, Insu Yun
   Registration date: 10-2803138-0000
   Patent number: 2025.04.28
   Country: Korea

1. **Security analysis system based on negative testing for protocol implementation of LTE device (Pending)**
   Inventors: Yongdae Kim, CheolJun Park, Sangwook Bae, Beomseok Oh, Jiho Lee, Mincheol Son, Insu Yun
   Application date: 2021.10.07
   Application number: 10-2021-0133149
   Country: Korea

Invited Talks

International

Title: Atlantis: an Autonomous LLM-Powered Bug Finding and Fixing System
- Presented at KAIST-NYU Workshop — Nov. 2025

Title: How to build Skynet — a system that hacks systems
- **Keynote** speech at TyphoonCon, Seoul, Korea — Jun. 2023

Title: HardsHeap: A Universal and Extensible Framework for Evaluating Secure Allocators
- Presented at ACM CCS 2021, Online — Nov. 2021

Title: Automatic Techniques to Systematically Discover New Heap Exploitation Primitives
- Presented at USENIX Security 2020, Online — Aug. 2020

Title: QSYM: A Practical Concolic Execution Engine Tailored for Hybrid Fuzzing
- Presented at USENIX Security 2018, Baltimore, MD — Aug. 2018

Title: APISan: Sanitizing API Usages through Semantic Cross-checking
- Presented at USENIX Security 2016, Austin, TX — Aug. 2016

Domestic

Title: The Singularity is Here: When AI Meets Cyber Security
- Seminar at Tech-Share by Samsung Electronics, Hyundai Motor, and Naver, Seoul, Korea — Feb. 2026

Title: The Dark and Bright Sides of AI in Cybersecurity
- Seminar at Samsung Electronics DS, Seoul, Korea — Jan. 2026
- Seminar at AI Security Technology Seminar, Seoul, Korea — Dec. 2025
- Seminar at KAIST AI Graduate School Colloquium — Oct. 2025
- Seminar at KISA, Seoul, Korea — Oct. 2025
- Seminar at KISIA, Seoul, Korea — Oct. 2025
- Seminar at KAIST-Cyber Security Policy Program — Sep. 2025
- Seminar at Samsung SDS, Seoul, Korea — Sep. 2025
- Seminar at National Intelligence Service, Seoul, Korea — Sep. 2025

Title: AI that Changed the Cybersecurity Talent and Education
- **Keynote** at Cybersecurity Talent Development Symposium, Seoul, Korea — Dec. 2025
- Seminar at Cyber Security Policy Forum, Seoul, Korea — Dec. 2025

Title: The Future of Security Transformed by AI
- Seminar at ACDC (AI Cyber Defense Contest), Seoul, Korea — Dec. 2025

Title: AI that Changed the Battlefield
- **Keynote** at WhiteHat 2025, Seoul, Korea — Nov. 2025

Title: Atlantis: an Autonomous LLM-Powered Bug Finding and Fixing System
- Seminar at Cyber Security Society, Seoul, Korea — Nov. 2025

- Seminar at Cyber Summit Korea 2025, Seoul, Korea — Sep. 2025
- Seminar at Tech-Share by Samsung Electronics, Hyundai Motor, and Naver, Seoul, Korea — Aug. 2025
- Seminar at ENKI WhiteHat, Seoul, Korea — Sep. 2025
- Seminar at LIGNex1, Seoul, Korea — Sep. 2025
- Seminar at KCSCON, Seoul, Korea — Sep. 2025

Title: AIxCC: The Future of Cybersecurity with LLMs
- Seminar at ETRI, Daejeon, Korea — Nov. 2024
- Seminar at Moon Sool Graduate School of Future Strategy, Daejeon, Korea — Nov. 2024
- Seminar at Future Security Tech Forum, Jeju, Korea — Nov. 2024
- Seminar at National Security Research Institute (NSRI), Daejeon, Korea — Sep. 2024

Title: 2024 Security Strategy: Polarization
- Seminar at Defense Counterintelligence Command, Gwacheon, Korea — May. 2024

Title: Building Automated Hacking Systems
- Seminar at POSTECH, Pohang, Korea — Nov. 2023

Title: Trends in Security Vulnerabilities of Low Earth Orbit Satellites
- Presented at ETRI, Daejeon, Korea — Aug. 2023

Title: Academic Research from Offensive Research
- Presented at Samsung, Seoul, Korea — Aug. 2023

Title: Human-friendly binary analysis
- Presented at ETRI, Daejeon, Korea — Nov. 2023
- Presented at Korea Computer Congress (KCC), Seoul, Korea — Jun. 2023

Title: Exploit in the wild
- Presented at ETRI, Daejeon — Jun. 2023

Title: Hacking 101
- Presented at WISC, Seoul — Sep. 2022

Title: Attack and Defenses for Heap Vulnerabilities in 2022
- Seminar at ETRI, Daejeon — Apr. 2022

Title: Comparative Analysis of Baseband Software and Cellular Specifications for Finding Vulnerabilities
- Seminar at UNIST, Ulsan — May. 2023
- Seminar at Security@KAIST, Online — Jun. 2022
- Seminar at Cyber Operations Command, Seoul — Jun. 2022

Title: Scalable and Automatic Vulnerability Discovery Beyond Random Testing

Seminar at Seoul National University, Seoul, Korea, Mar. 2019

Title: Memory Allocator Security
- Presented at Best of Best (BoB), Seoul — Feb. 2023
- Presented at Computer System Society Conference (CSC), Pyeongchang — Feb. 2023
- Seminar at UNIST, Online — May. 2022
- Seminar at Yonsei university, Online — Apr. 2022
- Seminar at Sungkyunkwan university, Online — Apr. 2022
- Seminar at ETRI, Daejeon — Jan. 2022
- Seminar at National Security Research Institute (NSRI), Daejeon — Dec. 2021
- Seminar at Securty@KAIST, Online — Nov. 2021
- Seminar at KAIST GSIS, Online — Nov. 2021

Title: Browser Security: Hacking & Research
- Presented at Open Theori Research Seminar #6, Online — Dec. 2021
- Seminar at Hanyang University, Online — Nov. 2021
- Presented at KR Becks Meetup #1 by LINE, Online — Aug. 2021
- Seminar at Security@KAIST, Online — Jun. 2021

Grants

To summarize, 2.4B KRW is awarded, and my portion is 1.96B KRW.

19. **Intelligent multi-agent system for variant vulnerability detection in large-scale software** 26.03 – 27.02
    Agency/Company: NRF
    Money: 142M KRW
    Role: PI
18. **Design and implementation of novel fuzzing system for USB driver security** 25.08 – 26.05
    Agency/Company: Samsung Electronics
    Money: 82.5M KRW
    Role: PI
17. **SAT-HACK: Satellite cyber kill chain verification research** 25.07 – 27.02
    Agency/Company: LIGNex1
    Money: 275M KRW
    Role: PI
16. **Software safety research based on automated vulnerability pattern generation** 25.04 – 25.10
    Agency/Company: NSRI
    Money: 60M KRW
    Role: PI
15. **AIxCC: Automating real-world vulnerability discovery and remediation** 24.09 – 26.02
    Agency/Company: NYU
    Money: 99.9M KRW
    Role: PI
14. **Research on building an open source kernel security model** 24.04 – 24.10
    Agency/Company: NSRI
    Money: 54.5M KRW
    Role: PI
13. **Revisiting IoT threat models for smart cities and developing a vulnerability analysis system based on these models** 24.01 – 25.12
    Agency/Company: IITP
    Money: 400M KRW
    Role: PI
12. **Building a system to assist variant analysis for browsers** 23.06 – 24.05
    Agency/Company: NRF
    Money: 65M KRW
    Role: PI
11. **Generating a security model based on JavaScript intermediate language** 23.04 – 23.10
    Agency/Company: NSRI
    Money: 54.5M KRW
    Role: PI
10. **Verifying security threats in open-source operating systems** 23.04 – 23.10
    Agency/Company: NSRI

Money: 54.5M KRW
Role: PI

9. **Browser fuzzing with formal verification for cross architectures** 22.09 – 23.09
   Agency/Company: NRF
   Money: 110M KRW
   Role: PI

8. **Building test suites for validating vulnerability detection** 22.08 – 22.11
   Agency/Company: ETRI
   Money: 27.3M KRW
   Role: PI

7. **Generating a security model based on JavaScript security analysis** 22.04 – 22.10
   Agency/Company: NSRI
   Money: 54.5M KRW
   Role: PI

6. **Developing techniques for collection and integrated analysis of automotive systems through event-based experimental systems** 22.04 – 23.12
   Agency/Company: Dankook university
   Money: 300M $\times$ 0.5 KRW
   Role: Co-PI with Prof. Yujip Won

5. **6G security** 21.08 – 23.09
   Agency/Company: Samsung Electronics
   Money: 200M $\times$ 0.2 KRW
   Role: Co-PI with Prof. Yongdae Kim

4. **DRAM security** 21.07 – 24.06
   Agency/Company: Samsung Electronics
   Money: 180M $\times$ 0.2 KRW
   Role: Co-PI with Prof. Yongdae Kim

3. **Systematic and precise transformation of the Qualcomm Hexagon architecture into intermediate representations for binary analysis** 21.06 – 22.05
   Agency/Company: NRF
   Money: 46.7M KRW
   Role: PI

2. **Automatically generating a security model for discovering web browser vulnerabilities** 21.04 – 21.10
   Agency/Company: NSRI
   Money: 54.5M KRW
   Role: PI

1. **Developing a scalable cyber reasoning system (Start-up)** 21.02 – 24.12
   Agency/Company: KAIST
   Money: 150M KRW
   Role: PI

Advising and Mentoring

Ph.D. Students

- Youngmin Choi (Co-advising with Prof. Kyoungsoo Park)
- Haein Lee
- Sujin Han
- Junyoung Park
- Minwoo Baek
- Eunkyu Lee
- Donghyeon Kim
- Dongjun Lee

Master Students

- Kyeongmin Kim
- Juhyun Song
- Minwoo Jeong
- Gunhee Ahn

Alumni (Ph.D.)

- HyungSeok Han (Co-advised with Yongdae Kim), Postdoc at Georgia Tech — Ph.D. in 2023

Alumni (M.S.)

- Dong-uk Kim, 78ResearchLab — M.S. in 2025
- Dongok Kim, ENKI WhiteHat — M.S. in 2025
- Hyeon Heo, ENKI WhiteHat — M.S. in 2025
- Seunggi Min, Financial Security Institute — M.S. in 2025
- Wonyoung Kim, Samsung — M.S. in 2025
- Taisic Yun (Co-advised with Yongdae Kim), Theori — M.S. in 2024
- Wonyoung Jung, 78ResearchLab — M.S. in 2024
- Hyunsik Jeong (Co-advised with Yongdae Kim), S2W — M.S. in 2021